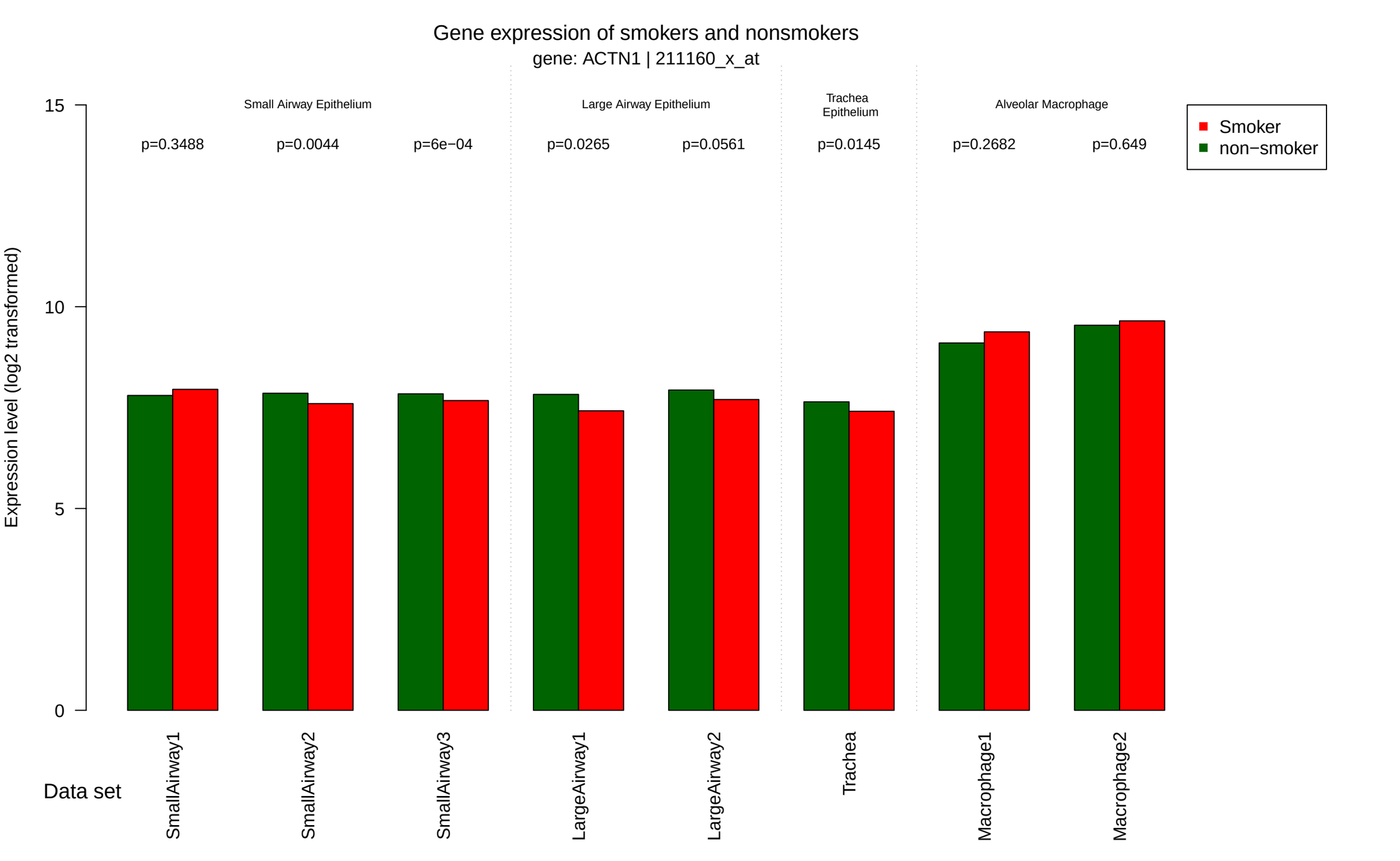
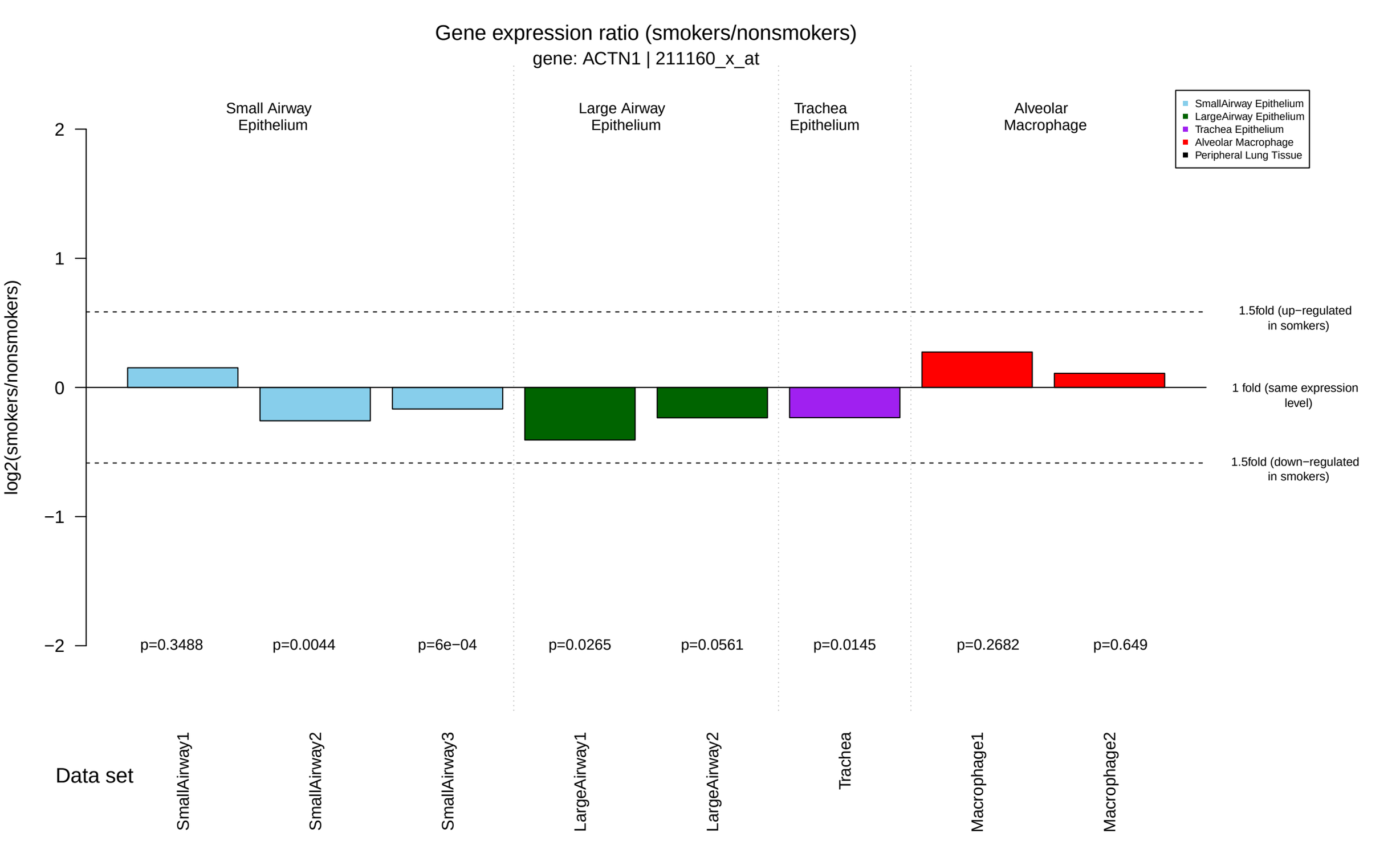
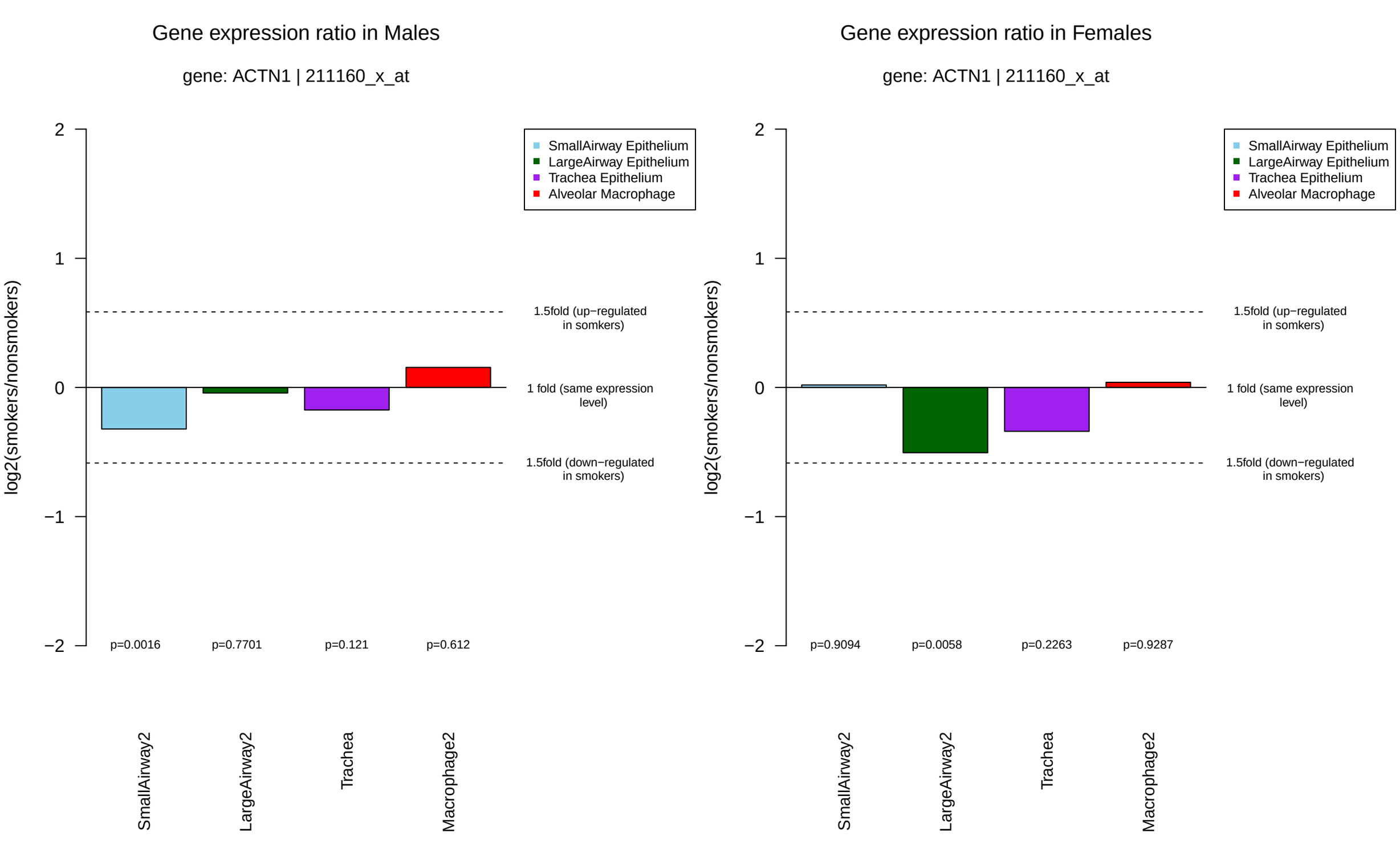
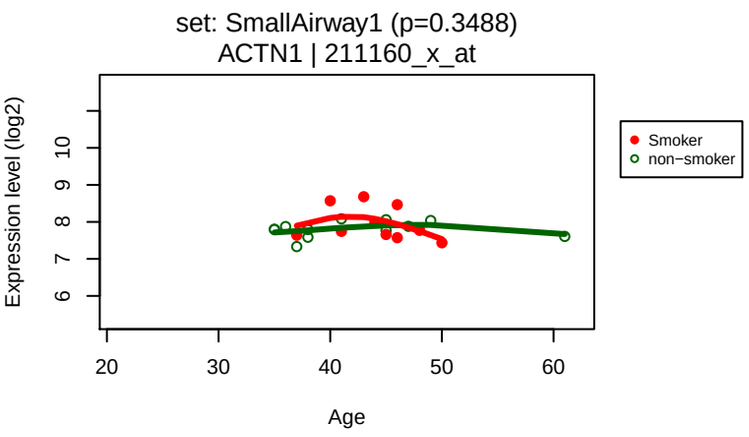
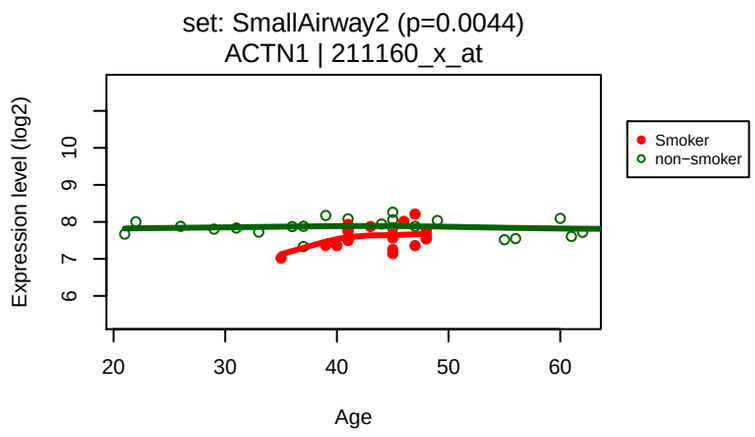
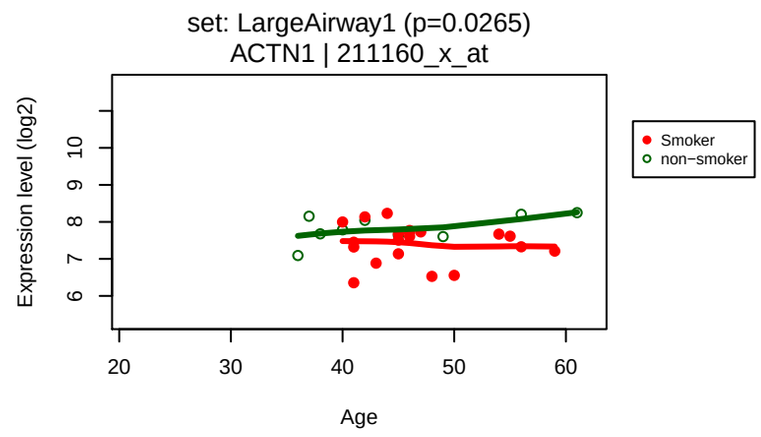

Gene expression of smokers and nonsmokers
gene: ACTN1 | 211160_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3488
p=0.0044
p=6e−04
p=0.0265
p=0.0561
p=0.0145
p=0.2682
p=0.649
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

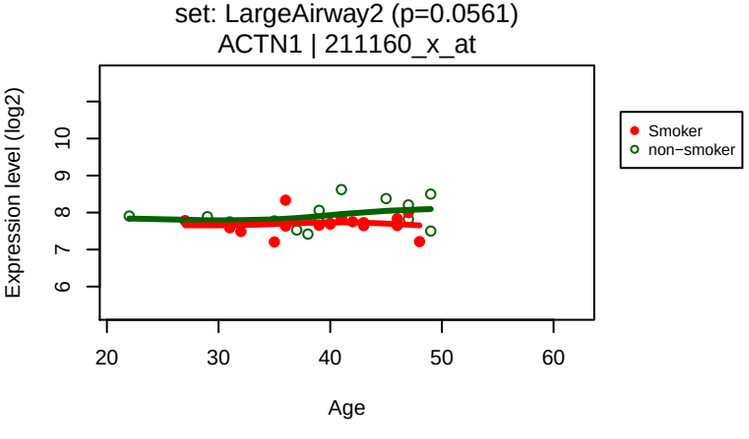

Gene expression ratio (smokers/nonsmokers)
gene: ACTN1 | 211160_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3488
p=0.0044
p=6e−04
p=0.0265
p=0.0561
p=0.0145
p=0.2682
p=0.649
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

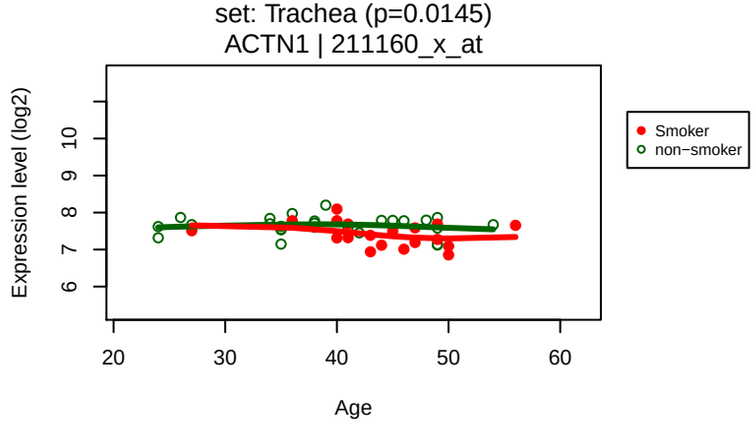

Gene expression ratio in Males
gene: ACTN1 | 211160_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0016
p=0.7701
p=0.121
p=0.612
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN1 | 211160_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9094
p=0.0058
p=0.2263
p=0.9287
SmallAirway2
LargeAirway2
Trachea
Macrophage2

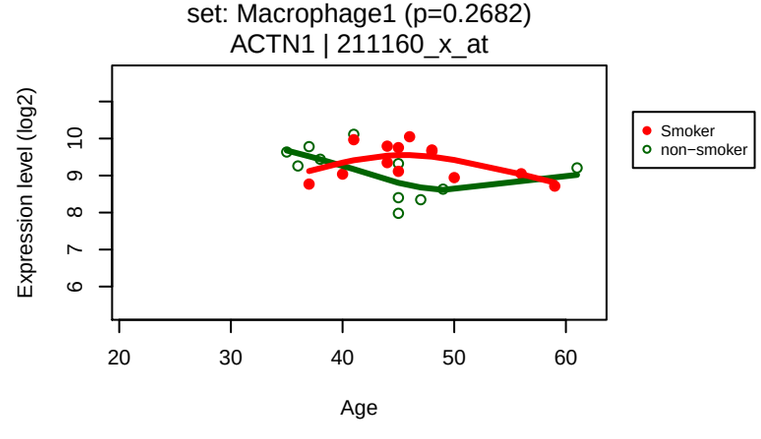

set: SmallAirway1 (p=0.3488)
ACTN1 | 211160_x_at
Expression level (log2)
Age
Smoker
non−smoker

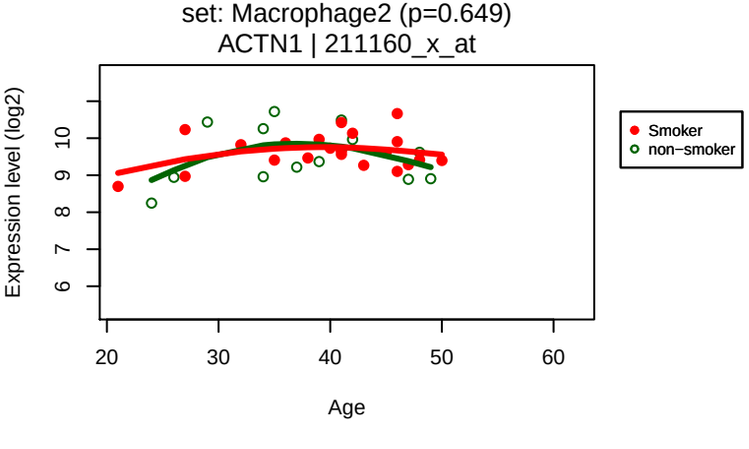

set: SmallAirway2 (p=0.0044)
ACTN1 | 211160_x_at
Expression level (log2)
Age
Smoker
non−smoker

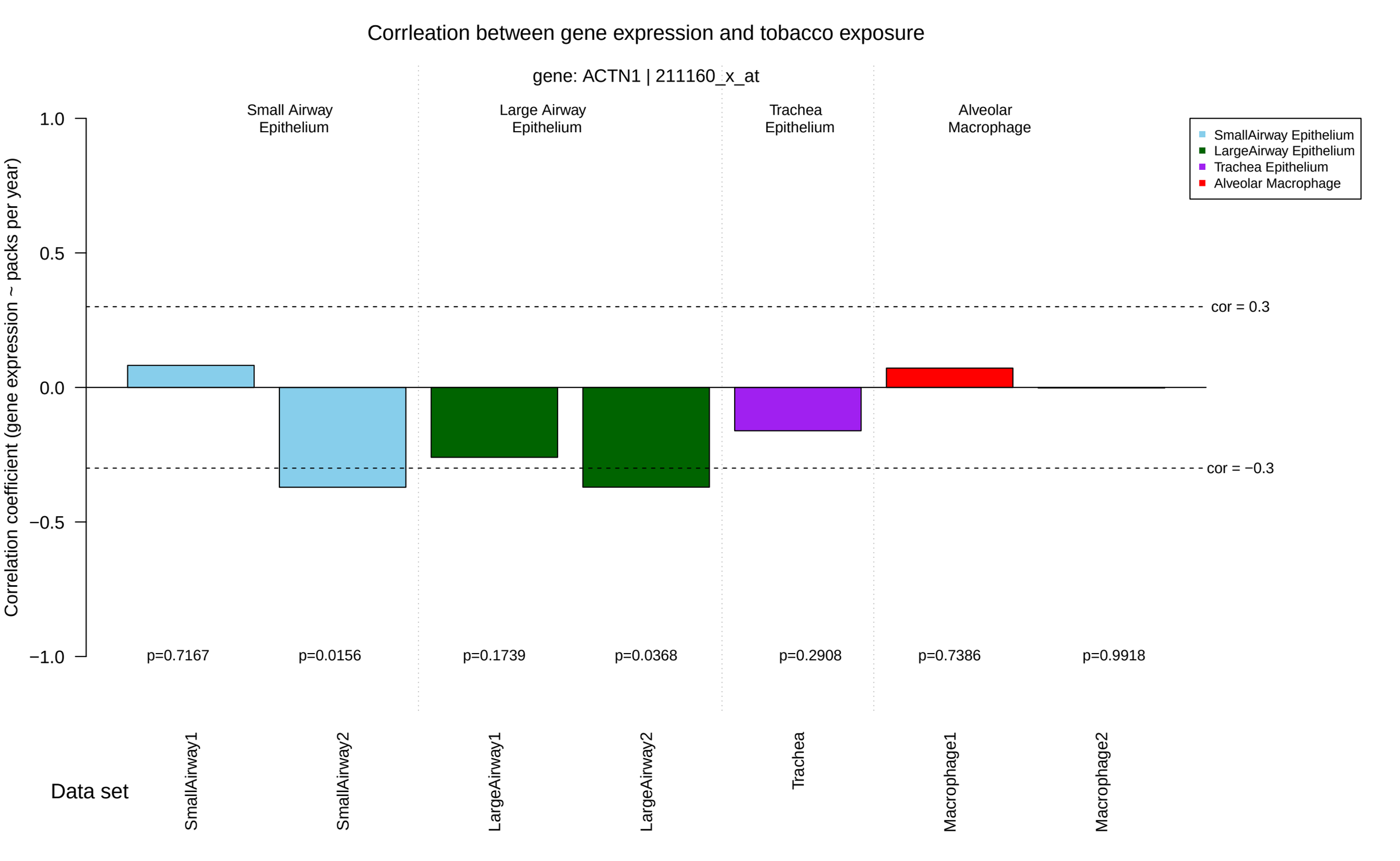

set: LargeAirway1 (p=0.0265)
ACTN1 | 211160_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0561)
ACTN1 | 211160_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0145)
ACTN1 | 211160_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2682)
ACTN1 | 211160_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.649)
ACTN1 | 211160_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN1 | 211160_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7167
p=0.0156
p=0.1739
p=0.0368
p=0.2908
p=0.7386
p=0.9918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2